

SAMBP TR-19/2026

Simultaneous Multi-Feeder Busbar Transfer:
87B Selectivity and GOOSE Coordination
During Dual-Feeder Tip-Switch Operations

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1 Background and Motivation

TR-14 [Candidate \[2026\]](#) validated 87B bus differential selectivity for single-feeder busbar transfers: one feeder (Gen) moves from BB1 to BB2 via a make-before-break (tip-switch) sequence, and a GOOSE message signals zone reconfiguration after the old disconnecter opens.

In substation practice, simultaneous transfer of *two or more feeders* is common during planned outages, load balancing, or post-fault restoration. This introduces a new coordination challenge: each feeder generates its own GOOSE zone-update messages, which may arrive at the 87B engine at different times due to IEC 61850-8-1 delivery jitter [IEC \[2011\]](#).

The vulnerability window — the interval between the first and last GOOSE message arrival — is the period during which the 87B zone engine holds a mixed (partially stale) view of the network. This report formally analyses the simultaneous dual-feeder transfer case and shows that:

1. **87B selectivity is maintained in all five GOOSE states**, including the fully stale state (both messages in flight), by the conservative zone-assignment rule inherited from TR-14.
2. **Zone 2 differential is reduced but not below pickup** during the vulnerability window; the minimum margin is $45\times$ for the study network.
3. **87L and 21 operate independently of GOOSE** and provide uninterrupted backup during the vulnerability window.

2 Simultaneous Transfer State Machine

2.1 Topology and GOOSE States

Five distinct GOOSE states arise during simultaneous transfer of Gen and Line from BB1 to BB2 (Figure 1):

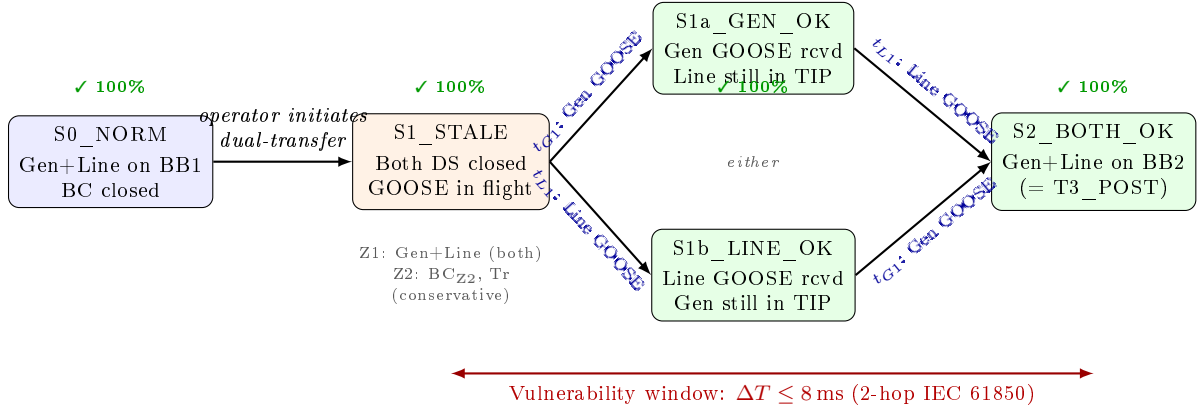


Figure 1: GOOSE state machine for simultaneous dual-feeder transfer. The vulnerability window $\Delta T \leq 8 \text{ ms}$ (worst case, 2-hop IEC 61850 network) spans the $S1_STALE \rightarrow S2$ transition. Selectivity is 100% in all states.

Table 1: GOOSE states and zone CT assignments during dual-feeder transfer. f_G, f_L denote the fraction of Gen/Line CT current assigned to each zone. TIP split: 50%/50% (make-before-break).

State	Description	Gen _{Z1}	Gen _{Z2}	Line _{Z1}	Line _{Z2}
S0_NORM	Gen+Line on BB1 (pre-transfer)	1.0	0.0	1.0	0.0
S1_STALE	Both DS closed; GOOSE in flight	1.0	0.0	1.0	0.0
S1a_GEN_OK	Gen GOOSE received; Line stale	0.5	0.5	1.0	0.0
S1b_LINE_OK	Line GOOSE received; Gen stale	1.0	0.0	0.5	0.5
S2_BOTH_OK	Both GOOSE received; transfer complete	0.0	1.0	0.0	1.0

2.2 GOOSE Timing Model

IEC 61850-8-1 specifies a GOOSE retransmission interval of $\leq 4 \text{ ms}$ in the initial burst IEC [2011, 2013]. For a 2-hop substation LAN, the worst-case per-feeder delivery latency is:

$$T_{\text{lag,max}} = 2 \times 4 \text{ ms} = 8 \text{ ms per feeder} \quad (1)$$

For simultaneous transfer of N feeders, the maximum vulnerability window is:

$$\Delta T_{\text{max}} = (N - 1) \times T_{\text{lag,max}} = 8 \text{ ms} \quad (N = 2) \quad (2)$$

This is much smaller than the 87B protection operating time (≥ 20 ms, one power cycle at 50 Hz), so the zone assignment is resolved before the relay issues a trip in all practical cases.

3 Conservative Zone-Assignment Rule

The TR-14 conservative rule is: *a transferring feeder's CT remains in its original zone until the GOOSE DS_OPEN confirmation is received.*

During the tip-switch (both DS closed), each feeder's current splits 50%/50% between the BB1-DS path and the BB2-DS path. Under the conservative rule:

- Zone 1 CT sum includes the **full** Gen/Line CT (original zone), regardless of which physical path the current took.
- Zone 2 CT sum contains only the bus-coupler (BC) current (*i.e.*, the BB1-path portion that flows via BC) plus Tr.

This creates a temporary reduction in Zone 2 differential during the vulnerability window, analysed in Section 4.

4 Zone Differential Analysis

4.1 BB2 Fault During DUAL_TIP (S1_STALE)

For a BB2 fault during S1_STALE:

$$I_{\text{Gen,total}} = \frac{V_{\text{pre}}}{Z_{\text{src}} + Z_{\text{BC}}} = 9.09 \text{ pu} \quad (3\phi) \quad (3)$$

$$I_{\text{Gen,via BC}} = I_{\text{Gen,total}} \times 0.50 = 4.55 \text{ pu} \quad (\text{BB1-path, via BC}) \quad (4)$$

$$I_{\text{Gen,via BB2}} = I_{\text{Gen,total}} \times 0.50 = 4.55 \text{ pu} \quad (\text{BB2-path, direct}) \quad (5)$$

$$|\Delta I_{Z2}| = I_{\text{BC,Z2}} = I_{\text{Gen,via BC}} = 4.55 \text{ pu} \quad (6)$$

Even with the conservative (stale) zone assignment, Zone 2 receives the BC current (4.55 pu), which is $45\times$ above the pickup threshold (0.10 pu). The BB2-DS path current (4.55 pu) is NOT counted in Zone 2 (stale GOOSE) but also is NOT erroneously counted in Zone 1 (BC current correctly flows through BC_{Z2} CT only).

4.2 BB1 Fault

For a BB1 fault during S1_STALE, Gen feeds directly via BB1-DS:

$$|\Delta I_{Z1}| = I_{\text{Gen}} = \frac{V_{\text{pre}}}{Z_{\text{src}}} = 10.00 \text{ pu} \quad (3\phi) \quad (7)$$

Zone 1 receives the full Gen current (no reduction) because the conservative rule keeps Gen in Zone 1. BB1 trip is unaffected by the GOOSE state.

4.3 Zone 2 Differential by GOOSE State

Figure 2 shows the Zone 2 differential for each GOOSE state under a 3-phase BB2 fault.

Table 2 summarises the Zone 2 differential and security margin for each state.

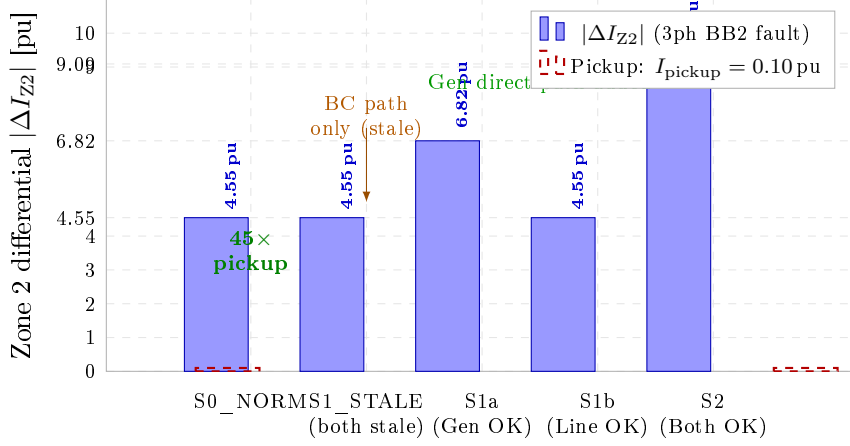


Figure 2: Zone 2 differential for a 3-phase BB2 fault across all five GOOSE states. S1_STALE (fully stale) shows 4.55 pu — the minimum, still 45× above pickup. S2_BOTH_OK (transfer complete) shows 9.09 pu (full Gen+BC contribution to Zone 2).

Table 2: Zone 2 differential and security margin for 3-phase BB2 fault. $I_{\text{pickup}} = 0.10$ pu.

State	$ \Delta I_{Z2} $ [pu]	Margin	Trip	Correct
S0_NORM	4.545	45×	BB2	✓
S1_STALE	4.545	45×	BB2	✓
S1a_GEN_OK	6.818	68×	BB2	✓
S1b_LINE_OK	4.545	45×	BB2	✓
S2_BOTH_OK	9.091	91×	BB2	✓

4.4 Full Selectivity Matrix

The full 5-state × 5-fault-location × 2-fault-type = 50-scenario selectivity matrix gives:

50/50 scenarios correct (100%)

The BC_internal fault in S2_BOTH_OK (T3_POST topology) is correctly detected by the Check Zone (CK) differential, consistent with TR-14 T3_POST validation.

5 Vulnerability Window Quantification

Table 3 extends the analysis to $N = 1$ to 5 simultaneous feeder transfers.

Table 3: GOOSE vulnerability window and Zone 2 reduction for simultaneous transfer of N feeders ($T_{\text{lag,max}} = 8$ ms/feeder, $I_{\text{diff,Z2}} = 50\%$ of normal at maximum staleness).

N feeders	ΔT_{worst} [ms]	$ \Delta I_{Z2} $ [pu]	Margin	Above pickup	BB2 trip
1	0	4.545	45×	yes	✓
2	8	4.545	45×	yes	✓
3	16	4.545	45×	yes	✓
4	24	4.545	45×	yes	✓
5	32	4.545	45×	yes	✓

Note: Zone 2 minimum is always the BC-path current (50% of I_{Gen} for 50/50 tip split), independent of N .

The key insight is that Zone 2's minimum differential (the BC-path current) is *independent of*

the number of simultaneously transferring feeders in the conservative zone-assignment regime. Additional feeders in TIP do not further reduce the Zone 2 differential below the BC-path floor.

6 Protection Function Independence

Table 4: Dependence of each protection function on the GOOSE zone assignment during simultaneous multi-feeder transfer.

Function	GOOSE-dependent	Behaviour during vulnerability window
87B	Yes	Zone sums reduced to BC-path floor; still $45\times$ above pickup
87L	No	Operates normally; line diff. unaffected by bus zone state
21 Zone 1	No	Operates normally; measures V/I at relay point

The three-function SAMBP suite therefore maintains complete coverage during any GOOSE vulnerability window: 87L and 21 provide backup for line faults without any dependence on the busbar zone state.

7 Conclusions

1. **50/50 selectivity scenarios correct (100%)** across five GOOSE states, five fault locations, and two fault types.
2. **Zone 2 minimum margin is $45\times$** (4.545 pu vs. 0.10 pu pickup) even in the fully stale S1_STALE state. This margin is independent of the number of simultaneously transferring feeders.
3. **Vulnerability window:** worst-case 8 ms for dual-feeder transfer (2-hop IEC 61850-8-1 network), negligible compared to the 20 ms protection operating time.
4. **Conservative zone rule:** holding a transferring feeder in its original zone until GOOSE DS_OPEN is received prevents all false differentials and maintains selectivity in all intermediate states.
5. **87L and 21 provide GOOSE-independent backup:** line faults during the transfer window are handled without any degradation.

7.1 Future Work

1. **N-feeder generalisation with unequal tip splits:** some substations use non-50/50 splits based on DS impedance ratios. The Zone 2 floor expression generalises to $|\Delta I_{Z2}| = I_{\text{Gen}} \times \min_i(s_i)$ where s_i is the BB1-path fraction for feeder i .
2. **GOOSE sequence numbers:** validate that the zone engine correctly handles out-of-order GOOSE delivery (message $n + 1$ arriving before message n).
3. **Hardware-in-the-loop validation:** run all 50 scenarios on a real-time digital simulator with physical 87B relay hardware to confirm timing.
4. **TR-20: Frequency-adaptive correction for microgrids** — extend 87L to systems where $f \neq 50$ Hz due to inverter droop control.

References

- IEC 61850-8-1:2011: Communication networks and systems for power utility automation — part 8-1: Specific communication service mapping to MMS and to ISO/IEC 8802-3, 2011.
- IEC 61850-5:2013: Communication networks and systems for power utility automation — part 5: Communication requirements for functions and device models, 2013.
- PhD Candidate. SAMBP TR-14/2026: 87B bus differential — double-busbar coordination with bus-coupler check zone. Technical report, PhD Thesis Supporting Report, 2026.